

Magnetact Animals: A Simple Kinetic Toy Kit for a Creative Online Workshop for Children

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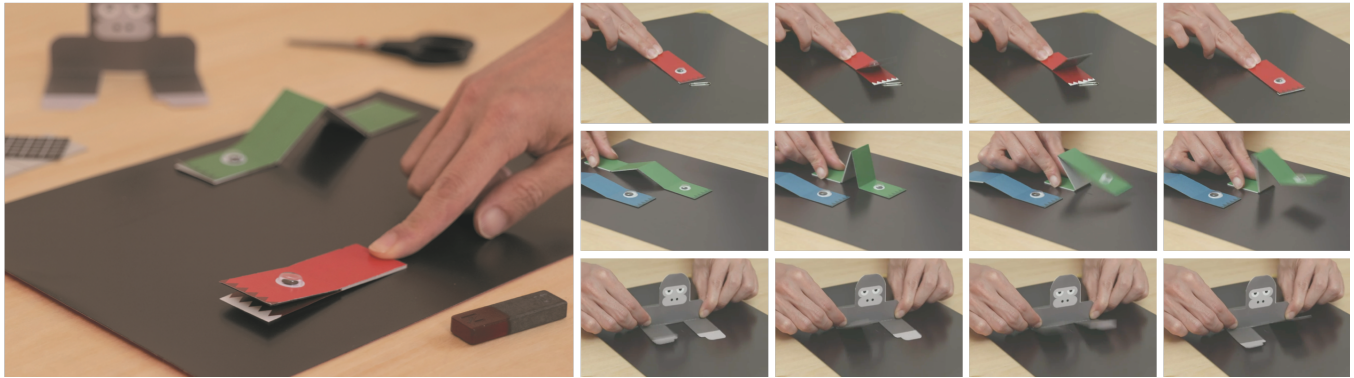


Figure 1: Magnetact Animals. A kinetic toy kit that can be sent in an envelope, be constructed with ease, and work without any batteries.

ABSTRACT

With the global COVID-19 pandemic that began in early 2020, it has become difficult to hold workshops that bring people together in one place. Consequently, demand for online events and virtual workshops is increasing so as to minimize the loss of learning opportunities. However, providing an experience of building things remotely is not easy because of the time and cost of preparing materials and equipment, and the difficulties of distance teaching. So, we designed a kinetic toy kit that can be sent in an envelope, be constructed with ease, and work without any batteries. The kinetic toy, constructed using magnetic sheets, cardboard, and paper, allows the users to design and create “animals” with variable motions. At the two online workshops we held for 15 participants from 3 to 11 years old, all of the children enjoyed building toys and some participants invented original mechanisms and new animals. In this paper, we describe the details of the toy kit and the online workshops that used it.

*Both authors contributed equally to this research.

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CCS CONCEPTS

• **Social and professional topics** → Children; • **Human-centered computing** → Collaborative interaction; **Interaction design**; • **Applied computing** → **Distance learning**.

KEYWORDS

Remote Workshop, Online Workshop, Magnet, Paper, Creative Toolkit

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1 INTRODUCTION

The global COVID-19 pandemic [2], which began in early 2020, has wholly changed our lifestyles and work styles. According to the guidelines issued by the World Health Organization (WHO), crowded places, close-contact settings, and confined and enclosed spaces should be avoided, as they increase the risk of infection [1]. Therefore, many schools and universities have been forced to close [11, 12].

Also, many workshop programs that have provided children with opportunities to experience the joy of creation and the fun of science, have been forced to consider changing their format to prevent the spread of the infection [13]. Hence, many creative communities [3, 5, 6], science museums [4, 7, 8], and companies [9]

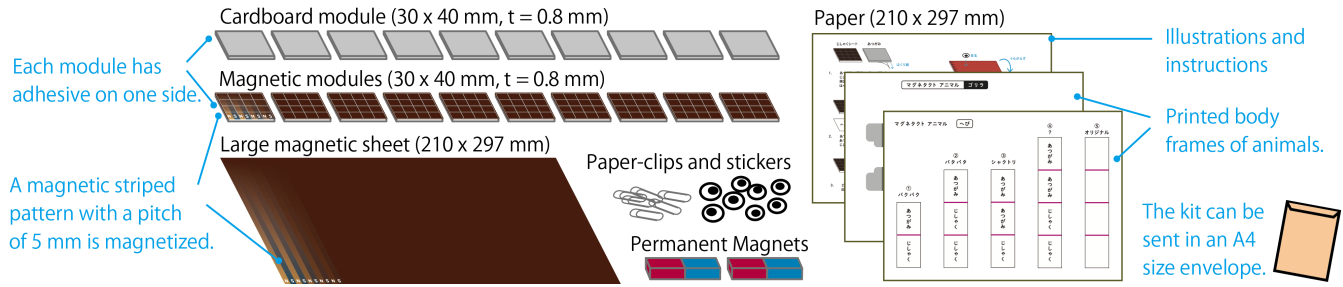


Figure 2: The kit includes magnetic sheet modules, cardboard modules, instruction papers, permanent magnets, some paper-clips, and stickers.

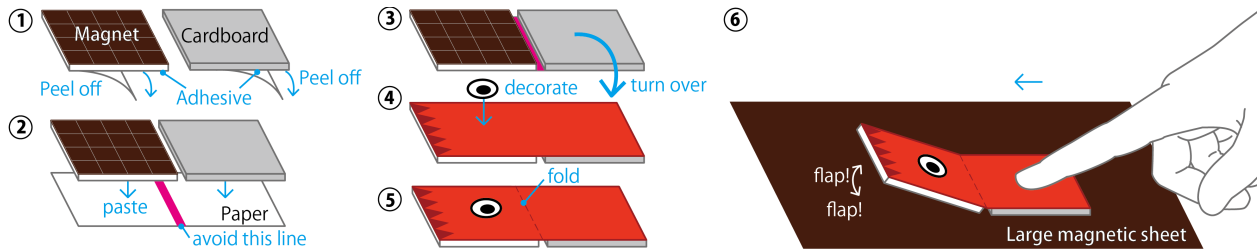


Figure 3: The basic construction of Magnetact Animals. Just a few steps are required to make a kinetic toy.

have published various types of online learning programs so that learning opportunities will not be lost during this outbreak period.

However, when facilitating an online creative workshop where participants use their hands to craft something, there are a number of hurdles for facilitators and participants to overcome.

First, unlike a normal creative workshop, each participant has to prepare the materials and tools by him/herself at his/her home in advance. Therefore, there are some creative programs shown that use only things likely to be available at home, such as paper, glue, scissors, and baking soda [3]. Alternatively, especially in workshops where the theme is robotics and electronics, there are several workshop programs where participants can receive a prepared kit in advance [6]. For such programs, many easy-to-assemble kits that have been proposed in the Human-Computer Interaction (HCI) research area might be used [10, 14, 16]. For example, TOPOBO [14] is a kit of blocks that can memorize movements, and which can be combined together to make a robot; LittleBits [10] enables kids to make electric circuits by combining several parts. Another example, Legoons [16], is a series of air-actuated blocks that can be combined with LEGO® blocks. These creative kits are very simple and do not require soldering and screwing for actuation. However, considering that the device cost and shipping charges will be added to the participation fee, workshops using inexpensive materials, with a less bulky, and simple structure might be easier to participate in.

Therefore, we propose a kinetic toy that can be easily sent, constructed, and animated (Figure 1), and propose an online creative workshop using it.

2 MAGNETACT ANIMALS

We named the kinetic toy “Magnetact Animals”. “Magnetact” [18] is a magnetic-sheet-based rapid prototyping method for haptic interfaces. By changing the magnetic pattern of the magnetic sheet using a handheld magnetizing tool, the magnetic force-based tactile feedback can be customized. By applying Magnetact, we designed a kinetic toy with not only tactile feedback but also movement and sound.

The materials required for the basic creation of Magnetact Animals are magnetic sheets, cardboard, and paper. They are all inexpensive materials which take the form of sheets, and therefore can be sent in an envelope (Figure 2). In addition, the animals can be processed with stationery such as scissors, glue, and tapes, and do not require any special tools. In the kit, pieces of magnetic sheet and cardboard are modularized to the same size and thickness. Each module has adhesive on one side, so it can be attached to the paper. This allows the workshop participants to easily create an animal-like structure. For the magnetic sheet module, referring to Magnetact [18] and Magnetic Plotter [17], we magnetized a magnetic striped pattern with a pitch of 5 mm using cylindrical neodymium magnets with a surface magnetic flux density of 382 mT.

For animal construction, illustrations and instructions are printed on the paper included with the kit. And the paper itself can be used to make the frames of the animals. By attaching the magnetic modules and cardboard modules to the paper, the paper connects the modules into a single unit (Figure 3).

For animal motion design, the kit includes a large magnetic sheet which has a magnetic stripe with the same pitch as the small magnetic sheets. This is used as a base on which to place the animals.

When the animal is placed on the base magnetic sheet and moved horizontally by hand, the magnetic fields on the animal side and the base magnetic sheet conflict with each other, and the small magnetic sheet will be lifted by a repulsive force generated between the magnetic sheets. But the cardboard modules do not react to the magnetic forces. Thus, the paper that connects the modules acts like a joint, and as a result the repulsive force of the magnetic sheet gives the animal various movements. For example, it is possible to easily create a crocodile that flaps its mouth, a snake that moves like an inchworm, and a gorilla drumming on the ground (Figure 1). This simple construction process gives workshop participants the experience of animating what they have created in the first few steps.

The next step is for the child to modify the look with stickers, pens, and scissors, and build his/her own animal. Participants can decorate the surface of the animals by painting or drawing on it, or by applying stickers. Alternatively, more skilled participants can create different motions and new animals by changing the position and orientation of the magnetic modules, which affect the way the animal moves (Figure 4). This is just like embedding components in a program and running it. Since this kinetic mechanism and actuation process has a similar aspect to a physical programming kit, even participants who are good at making things can enjoy creativity and ingenuity. Thus, even if the participants' backgrounds and ages have a broad variance, the second step in the construction process can be customized according to the participants' skill. This design reflects the fact that one of the aims of this workshop kit is to spread knowledge to more participants, deepen their understanding, and get them interested in science and technology.

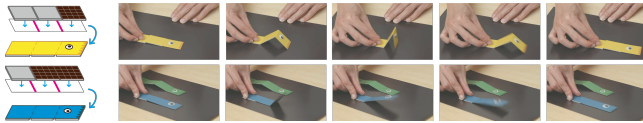


Figure 4: Even if the animals have the same body shape, their motions will change depending on their structure.

3 WORKSHOP

We have held two online creative workshops using this kit. Here are the details of the workshop.

3.1 Preparation

We planned 90-minute online workshops in the morning and afternoon of May 20, 2020. Participants were recruited from the families of employees of the NTT Science and Core Technology Laboratory Group with children aged 3 to 12 years old, and 15 participants from 3 to 11 years old were selected on a first-come, first-served basis. The table below shows the age distribution of participants (Table 1). Then, we mailed envelopes of the workshop kit to each participant.

3.2 Flow of the workshop

On the day of the workshops, a tutor, facilitators, and participants were connected using Cisco WebEx Meetings to share audio and

Table 1: Age and number of participants in each workshop.

Age	3	4	5	6	7	8	9	10	11	12
AM	1	0	1	2	0	2	2	0	1	0
PM	0	2	3	0	1	0	0	0	0	0

video (Figure 5). On the tutor side, three cameras were installed, providing a bust shot of the tutor, an overhead shot of the desktop, and a close-up shot of the tutor's hands. In online workshops, the information that can be sent is relatively limited compared to regular face-to-face workshops [15]. To compensate for this, we installed a multi-camera system for a lecture on how to make Magnetact animals. By switching between the three cameras, the tutor can explain each step from an appropriate angle.

Then, we held the online workshops with the following timeline.

- 5 minutes [Greetings]: At first, we made sure that participants had the tools and materials at hand.
- 5 minutes [Introduction video]: We played a video to share the final image of the toy we would create.
- 10 minutes [Explanation]: Without using presentation software, the tutor showed printed slides and some magnetic demonstrations from the desktop view, explaining the magnetic force and Magnetact Animals as if reading a picture book to the participants.
- 10 minutes [1st creation]: Participants had their first experience of making and moving the smallest animal.
- 20 minutes [2nd & 3rd creation]: Participants made two more animals and learned that the movement changes depending on the length of the body and the position of the attached magnetic modules.
- 25 minutes [4th creation]: The tutor gave a lecture on making the biggest Magnetact Animal, a "gorilla".
- 5 minutes [Summary]: Participants played with the animals they created and finished the workshop.

At the end of the workshop, we asked every participant to submit some photos and videos of the animals they made that day, or of animals improved or created within a period of seven days following the workshop.

3.3 Results of the Workshops

First, we were able to carry out both of the online workshops to the end without causing any participant to experience failure. This was thanks not only to the simple configuration of the Magnetact Animals kit, but also to the interest stimulated in participants by the video shared at the beginning of the workshop, and to the effectiveness of the multi-camera system in providing visual information from the optimum angle during the workshops.

In addition, one week after the workshops, 14 participants submitted videos and photos of their creative works. The created animals were decorated with paper and pens, and had a variety of unique shapes such as a snail, a frog, an insect, and other kinds of creatures (Figure 6). Furthermore, three of these participants (8, 9, 11 y/o) submitted new animal mechanisms such as fluttering birds and insects. The animals have wings of magnetic sheets and flap their wings when they are moved on the magnetic base. Even

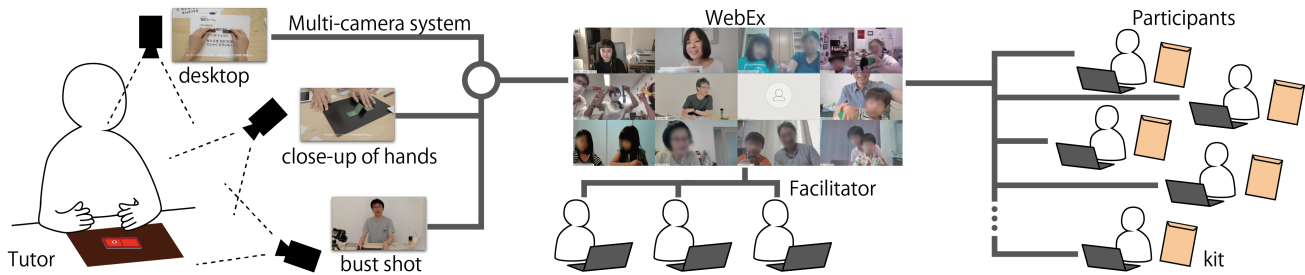


Figure 5: A tutor, facilitators, and participants were connected using WebEx Meetings. On the tutor side, a multi-camera system was installed. By switching between the three cameras, the tutor could explain each step from an appropriate angle.

though the tutor did not specifically teach such structures in the workshop, the participants were able to invent the new animals by themselves. This shows that the kit has high potential applicability and suitability for young creators with a wide range of skills. We expect to consider further diversification of workshops in the future.

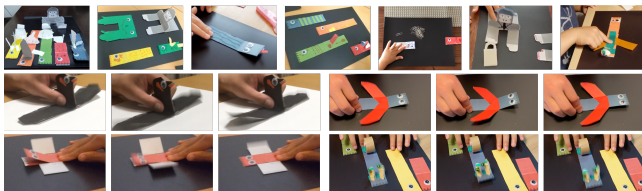


Figure 6: Workshop results. One week after the workshops, 14 participants submitted videos and photos of their creative works.

4 CONCLUSION

In this paper, we proposed a flat materials-based kinetic toy and an online workshop using it. The kinetic toys we designed using magnetic sheets and paper are made of inexpensive materials, can be sent in an envelope, and can be easily constructed. To try out this kit, we held two online workshops each attended by 15 participants aged from 3 to 11 years old. All the participants enjoyed the craft and some participants discovered original mechanisms by themselves. The results of the workshops show that it is possible to produce various animals and movements by changing the construction, and it was suggested that although the structure of this kit is very simple, it has high applicability.

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